NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

CS5224 CLOUD COMPUTING · FINAL PRESENTATION · DYLAN CHAN · XIN WEI CHONG · KELVIN LOW · JOSHUA LUM · BRANDON TAN

$10k+

to test one ad on a brain scanner

fMRI study	**$10 – 15k**	2–4 weeks
NeuroLens	**$1.00**	≈ 7 min

Ten cuts by lunch.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



What already exists: measured, filmed or guessed.

Brain scans (fMRI, EEG) research labs, iMotions suites	$10,000 – $15,000	2–4 weeks
Facial coding Realeyes, iMotions webcam studies	$850 – $4,000	3–5 days
AI attention maps Neurons AI, Attention Insight	$500 – $1,500 / mo	sub-minute
NeuroLens predicted brain response	$1.00	≈ 7 min

OUR DIFFERENCE

A brain response, not a gaze or a face

Per second, with a no-sound pass

Compare any two cuts for $1.00

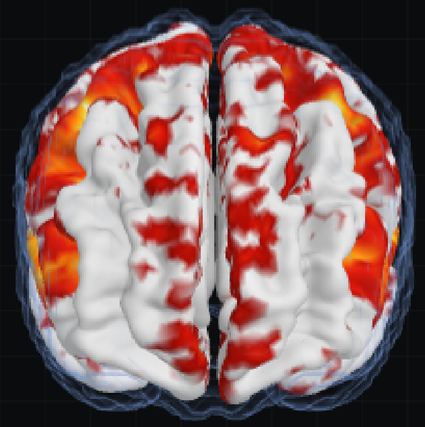
Under the hood: TRIBE v2.

Frames
V-JEPA 2 watches

Sound
Wav2Vec-BERT listens

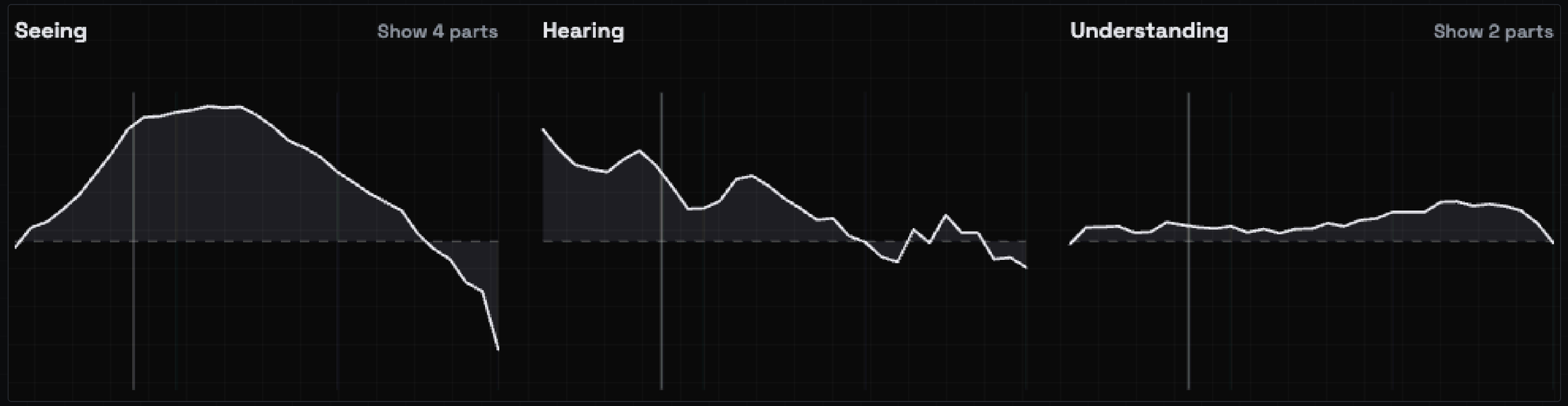
Words
WhisperX + LLaMA 3.2 read

TRIBE v2
1,000+ hours of fMRI
from 720 people



20,484 cortex points, every second

Seven signals, three questions.



SEEING

faces · bodies · scenes · visual

HEARING

sound

UNDERSTANDING

speech · social

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

Pay as you go. Credit, not cards.

$1.00

per started 30 s of video

NEW USERS: $10 CREDIT = TEN 30 S VIDEOS

WHY PREPAID CREDIT: CARD FEE AS A SHARE OF THE CHARGE

A $1.00 video, paid by card

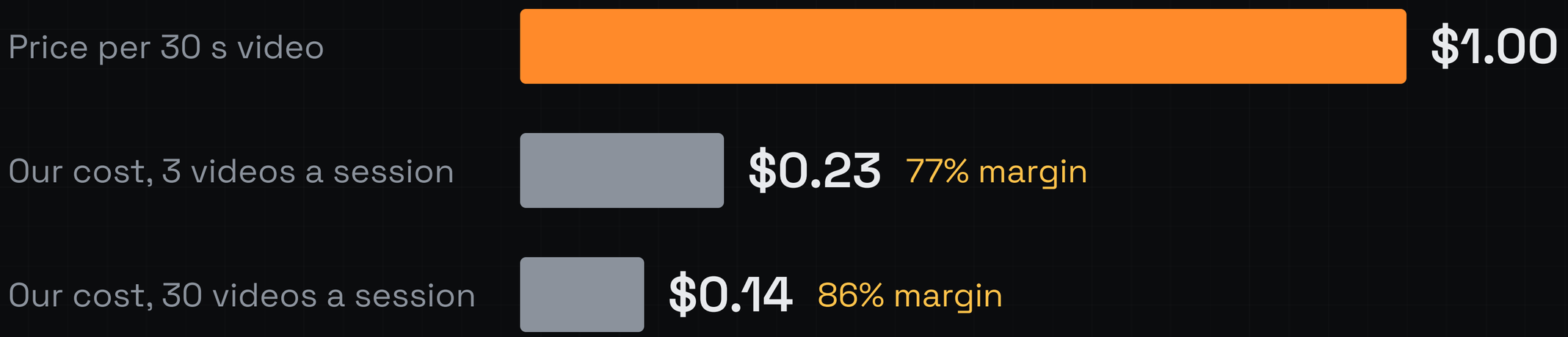
≈ 33%

A $20 credit pack

≈ 4%

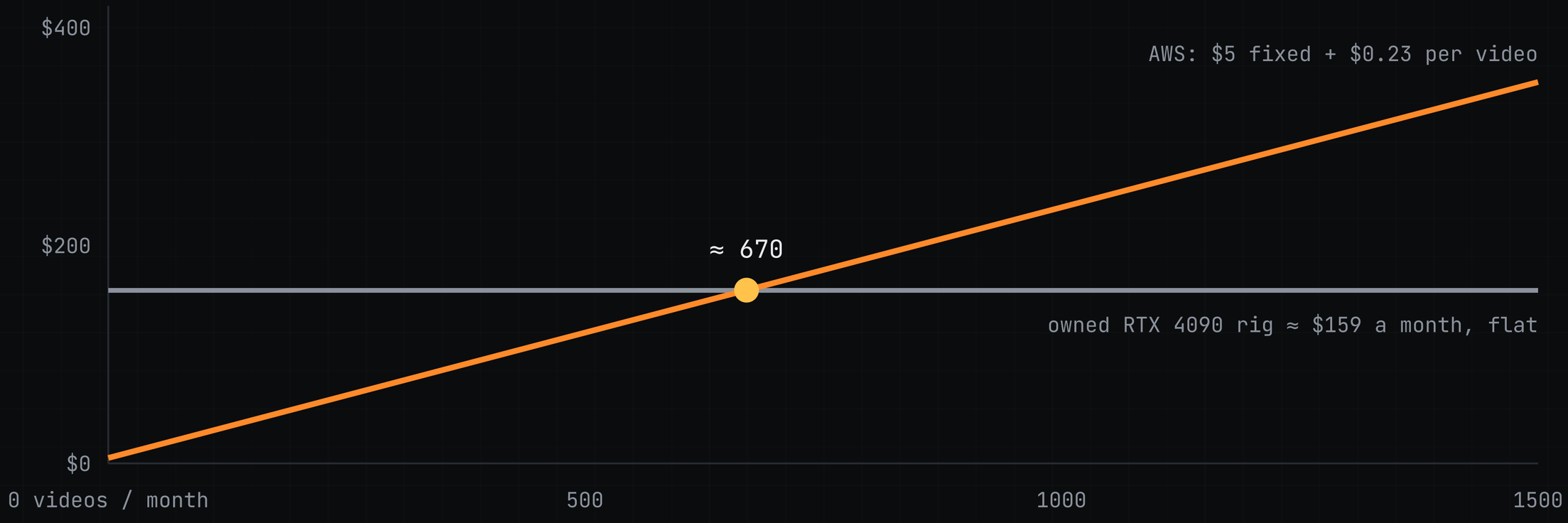
COURSE VERSION: AN ADMINISTRATOR GRANTS CREDIT · STRIPE CHECKOUT BELONGS TO THE PRODUCTION DESIGN

What one analysis costs us.

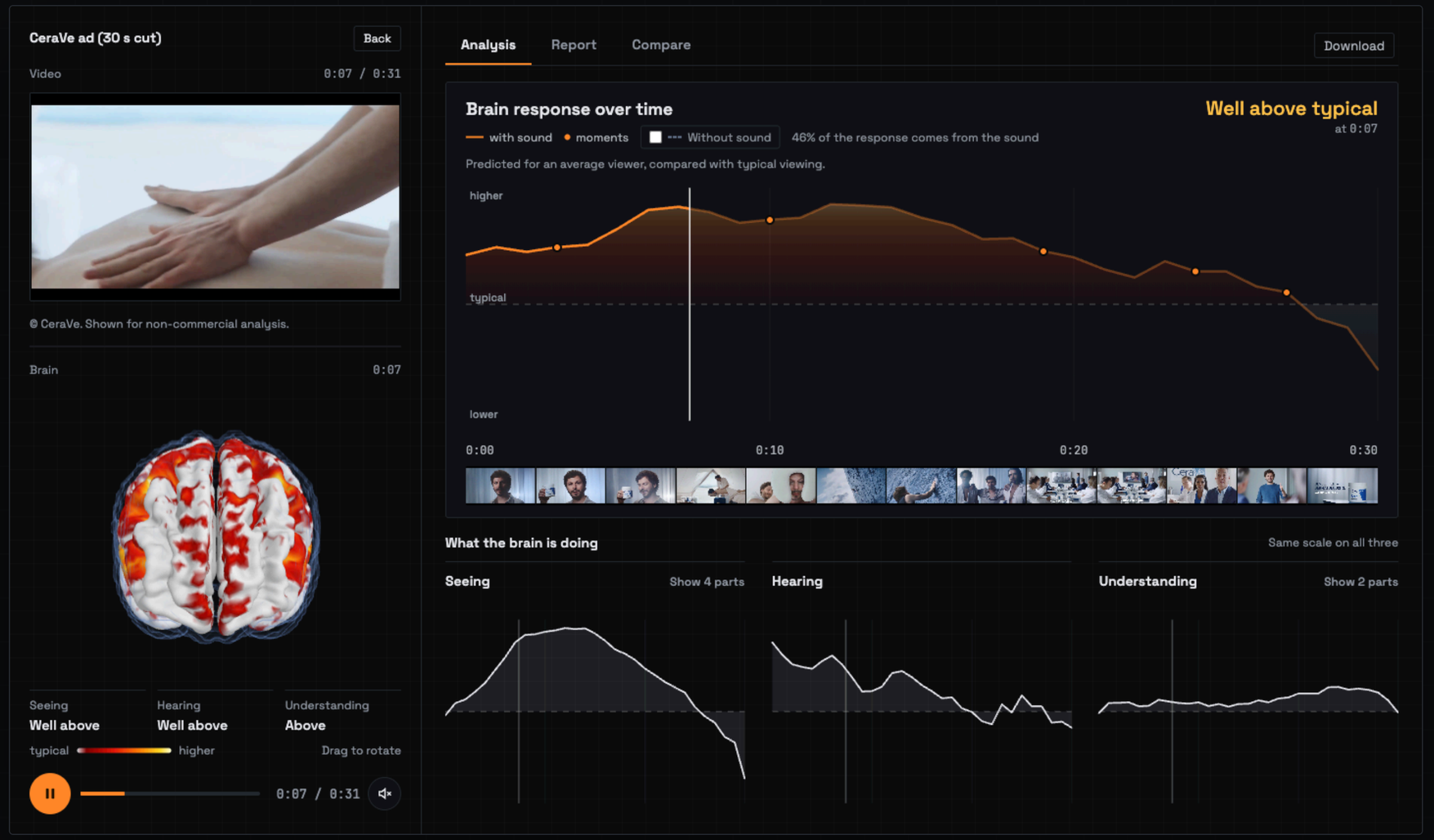


GPU 393 S AT $1.22/H (L4) · START-UP AND IDLE SHARED ACROSS A SESSION · FIXED COST $5 A MONTH

Cloud is cheaper below about 670 videos a month.



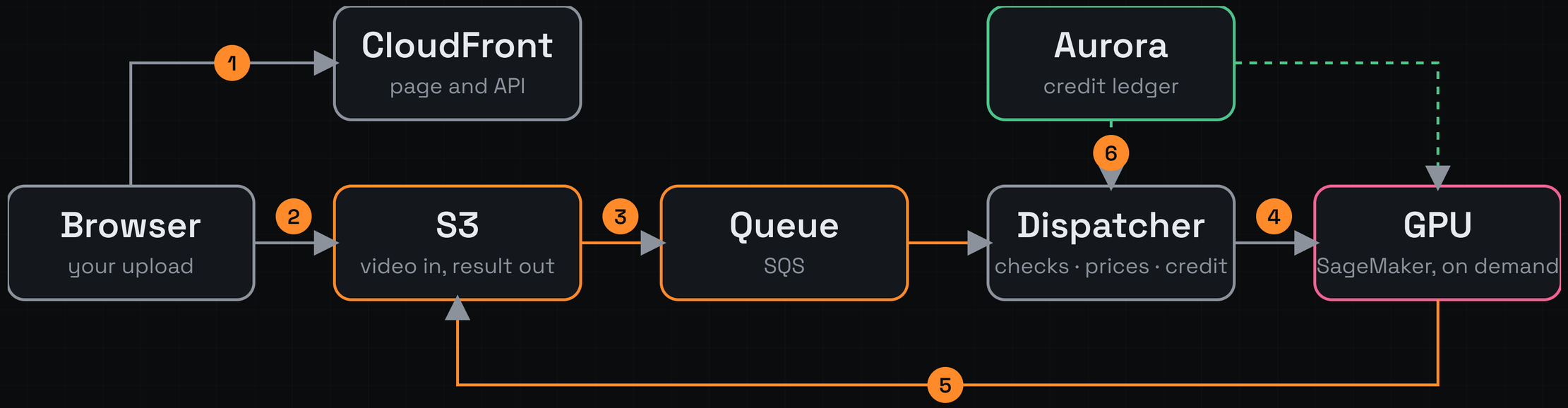
LIVE Live demo: sign in, upload, explore



1. Upload: price first
2. Peak, dip, no-sound toggle
3. Seeing · hearing · understanding
4. AI report, compare, bad file

CLICK THE PICTURE TO OPEN THE LIVE SITE · DF6GIRQYYKYFA.CLOUDFRONT.NET

How a video becomes a result.



ORANGE STORAGE AND QUEUE · PINK GPU · GREEN DATABASE · DASHED: CREDIT

Demo

We pay for GPU minutes only when someone is waiting.



Check first

Length and credit are verified before any GPU wakes.

Scale to zero

Off after 7 idle minutes: about $5 a month when idle.

Refund by default

Charged or refunded exactly once.

How do we know it works?

THE MODEL: VALIDATED BY META

720

people scanned; accurate on new videos, tasks and people. Not re-tested by us.

OUR PIPELINE

0.005

largest difference, RTX 4090 vs AWS, across 11 series

77/80

refund tests passed

OUR USERS

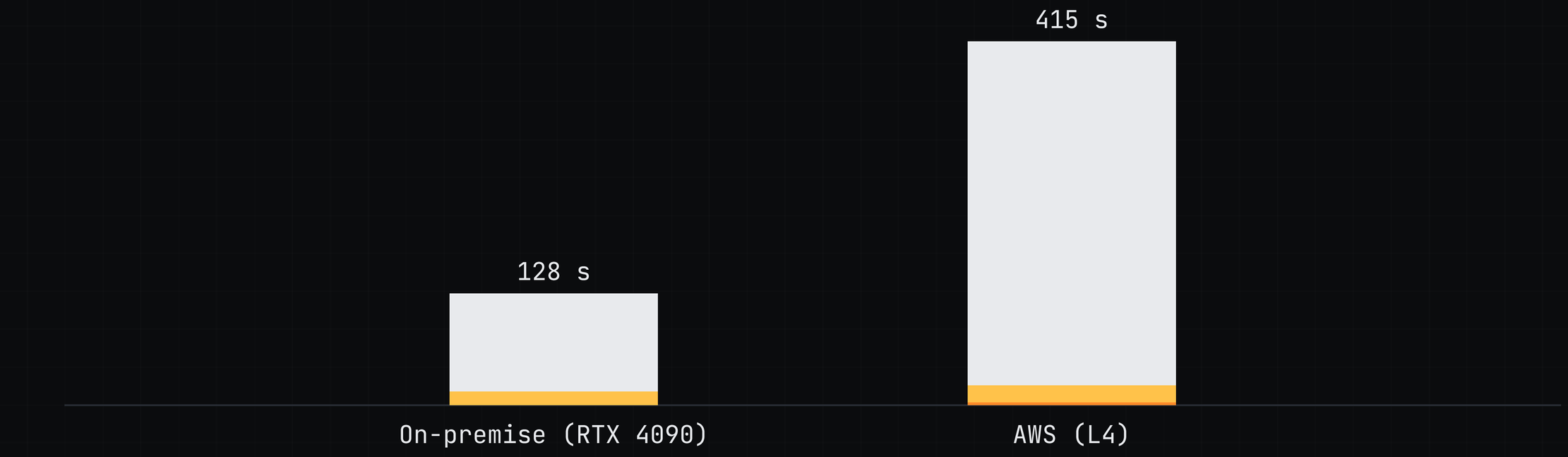
[TBD]

SUS score and survey: see Usability

Not validated: that a higher response means more views or sales.

Where does the time go in one job?

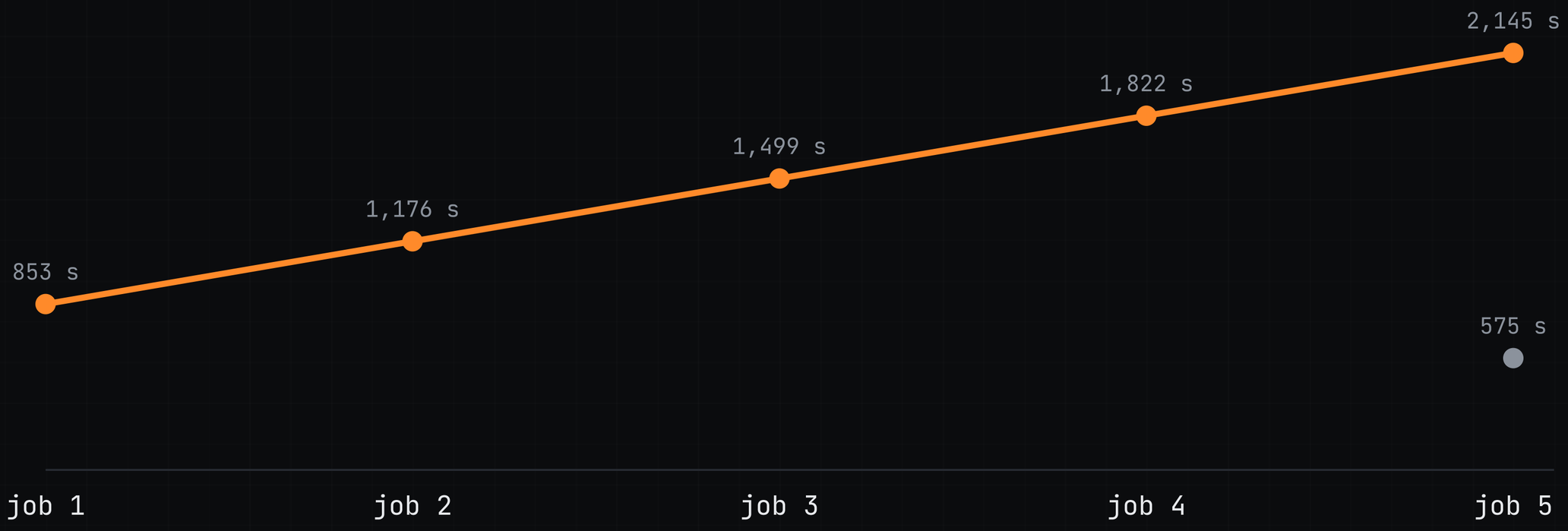
Same code, same 30 s clip, on our own RTX 4090 and on AWS.



GPU TIME PER VIDEO-SECOND: 3.71 S (RTX 4090) VS 10.7 S (A10G) · FIRST VIDEO AFTER 7+ MIN IDLE: 779 S ON AWS

Five people at once.

Five uploads at once. The endpoint is set to one GPU, so jobs run one after another.



SECONDS UNTIL EACH JOB FINISHES. NO JOB WAS DROPPED. THE AWS RUN GOT AN A10G, AND ITS FIRST JOB APPEARS TO HAVE RESTARTED ONCE.

Can people use it?

Booth visitors tried a sample or uploaded a clip, then answered the 10-question SUS.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS VERY GOOD

[X]% SAW SOMETHING THEY HAD NOT NOTICED
[Y]% WOULD PAY $1.00 FOR A 30 S VIDEO

What it claims, and what it doesn’t.

WHAT IT IS

- An average viewer’s brain response
- One scale for every video

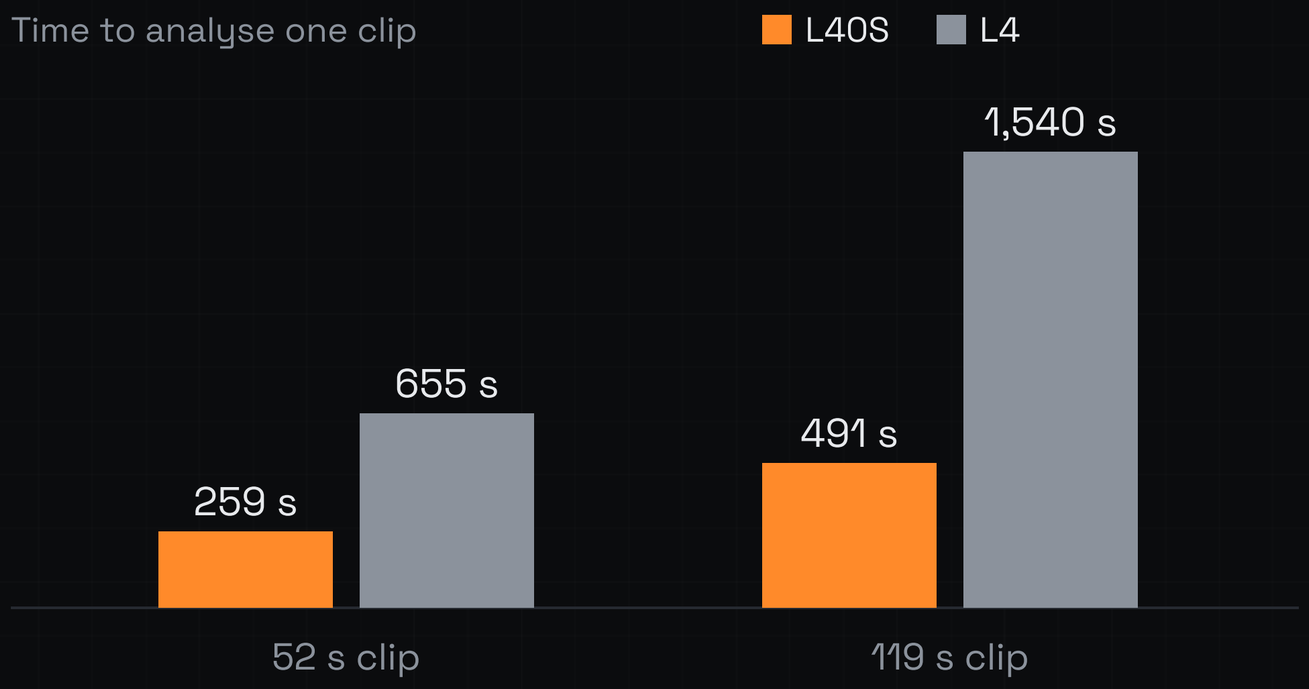
WHAT IT ISN'T

- A prediction of views or sales
- Instant: first video ≈ 6 min

ETHICS

- Non-commercial licence (CC BY-NC)
- No one scanned or profiled
- Anonymous, consented survey

Where next: scaling up.



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Spot launches failed for lack of capacity

19.9 GB

peak GPU memory: a 24 GB card fits

An EC2 GPU pool: $0.10 – $0.20 a video, about $70 more a month. It pays off above about 1,000 videos a month. Estimates, not built.

A $10,000 brain study, in 7 minutes for $1.

$1.00
per 30 s video

≈ 7 min
to a result, GPU warm

670
videos a month: break-even vs owning

≈ 6 min
first video after a quiet spell

Next

Licence

CC BY-NC: a paid launch needs one.

Next

Scale

A GPU pool that grows with the queue.

Next

Polish

Fix what our usability study flagged.

Thank you.
Questions?

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CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS